

## 0.1 Compactness

**Definition 0.1.1.** A Topological Space  $X$  is said to be *compact* if: Every open cover contains a finite subcover. i.e.,

$$\text{If } X = \bigcup_{\alpha \in \Lambda} \mathcal{U}_\alpha, (\mathcal{U}_\alpha \in \mathcal{T}), \text{ then there is finite subcover such that } X = \bigcup_{i=1}^N \mathcal{U}_{\alpha_i}$$

This is equivalent with:

$$\text{If } \emptyset = \bigcap_{\alpha \in \Lambda} \mathcal{C}_\alpha, (\mathcal{C}_\alpha \text{ closed}), \text{ then there is finite subset such that } \emptyset = \bigcap_{i=1}^N \mathcal{C}_{\alpha_i}$$

**Definition 0.1.2.** Let  $X$  be a set. We say that a collection  $\mathcal{A} \subset \mathcal{P}(X)$  satisfies *finite intersection property* if:

Every finite intersection of elements in  $\mathcal{A}$  is Not empty.

**Example.** There are collections which satisfies FIP, but infinite intersection is empty.

1.  $X = \mathbb{R}$ , and let  $\mathcal{A} = \{(n, \infty) \mid n \in \mathbb{N}\}$ .
2.  $X = \mathbb{R}$ , and let  $\mathcal{A} = \{\mathbb{R} \setminus F \mid F \text{ is finite set}\}$ .

**Theorem 0.1.3.** Let  $X$  be a Topological Space. TFAE:

- a)  $X$  is Compact Space.
- b) If  $\mathcal{A}$  is a collection of closed subsets of  $X$  satisfying FIP, then  $\bigcap_{C \in \mathcal{A}} C \neq \emptyset$ .
- c) If  $\mathcal{A}$  is a collection of subsets of  $X$  satisfying FIP, then  $\bigcap_{S \in \mathcal{A}} \overline{S} \neq \emptyset$ .

*Proof.* a)  $\implies$  b). Proof by Contradiction:

Suppose that  $\mathcal{A} \subset \mathcal{P}(X)$  is a collection of closed subsets such that FIP. Assume that  $\bigcap_{C \in \mathcal{A}} C = \emptyset$ .

$$\emptyset = \bigcap_{C \in \mathcal{A}} C \text{ if and only if } X = \bigcup_{C \in \mathcal{A}} (X \setminus C)$$

Since  $X \setminus C$  is open, there is a finite subcover by Compactness:

$$X = \bigcup_{i=1}^N (X \setminus C_i) \text{ if and only if } \emptyset = \bigcap_{i=1}^N C_i$$

This is Contradiction with  $\mathcal{A}$  satisfies FIP.

b)  $\implies$  a). Proof by Contraposition:

Suppose that  $X$  is not Compact. Then, there exists an Open Cover  $\mathcal{O}$  with no finite subcover: i.e.,

$$X = \bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U} \text{ if and only if } \emptyset = \bigcap_{\mathcal{U} \in \mathcal{O}} (X \setminus \mathcal{U})$$

And,

$$\text{For any finite subset of } \mathcal{O}, F = \{\mathcal{U}_i \mid i = 1, \dots, N\} \text{ satisfies } X \supsetneq \bigcup_{i=1}^N \mathcal{U}_i \text{ if and only if } \emptyset \neq \bigcap_{i=1}^N (X \setminus \mathcal{U}_i)$$

Thus,  $\mathcal{K} = \{X \setminus \mathcal{U} \mid \mathcal{U} \in \mathcal{O}\}$  satisfies FIP, but  $\emptyset = \bigcap_{\mathcal{U} \in \mathcal{O}} (X \setminus \mathcal{U}) = \bigcap_{C \in \mathcal{K}} C$ . Thus, not a) implies not b).

c)  $\implies$  b) is trivial. □

**Theorem 0.1.4.** Let  $X$  is Compact Space,  $Y$  is Topological Space.  
If  $f : X \rightarrow Y$  is Continuous Map, then  $f[X]$  is Compact.

*Proof.* Let  $\mathcal{O}$  be an open cover of  $f[X]$ . i.e,  $f[X] \subset \bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U}$ . Now,

$$X \subset f^{-1}[f[X]] \subset f^{-1} \left[ \bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U} \right] = \bigcup_{\mathcal{U} \in \mathcal{O} \text{ open, } f \text{ conti.}} f^{-1}[\mathcal{U}]$$

Since  $X$  is compact, there is a finite subcover such that

$$X \subset \bigcup_{i=1}^N f^{-1}[\mathcal{U}_i]$$

Consequently,

$$f[X] \subset f \left[ \bigcup_{i=1}^N f^{-1}[\mathcal{U}_i] \right] = \bigcup_{i=1}^N f[f^{-1}[\mathcal{U}_i]] \subset \bigcup_{i=1}^N \mathcal{U}_i$$

□

**Theorem 0.1.5.** Closed set of compact space is compact.

*Proof.* Let  $X$  be a compact, and  $E \subset X$  be a closed subset. Let  $\mathcal{O}$  be an open over of  $E$ . Then,

$$X = E \cup (X \setminus E) \subset \left( \bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U} \right) \cup (X \setminus E)$$

is an open cover of  $X$ . Thus, there is a finite subcover such that

$$X = \left( \bigcup_{i=1}^N \mathcal{U}_i \right) \cup (X \setminus E) \iff E \subset \bigcup_{i=1}^N \mathcal{U}_i$$

□

**Theorem 0.1.6.** Let  $X$  be a Topological Space, and  $\beta$  be a basis of  $X$ . Then, TFAE:

- a)  $X$  is Compact Space.
- b) Every open cover consisting of basis elements has a finite subcover.

*Proof.* a)  $\implies$  b). Clear by definition of Compact.

b)  $\implies$  a). Let  $\{\mathcal{U}_\alpha \mid \alpha \in \Lambda\}$  be an Open cover of  $X$ . That is,

$$X = \bigcup_{\alpha \in \Lambda} \mathcal{U}_\alpha = \bigcup_{\alpha \in \Lambda} \bigcup_{\gamma \in \Gamma_\alpha} B_\alpha^\gamma$$

where  $\{B_\alpha^\gamma \mid \gamma \in \Gamma_\alpha\}$  is subset of basis such that  $\bigcup_{\gamma \in \Gamma_\alpha} B_\alpha^\gamma = \mathcal{U}_\alpha$ . Now, by 2), there is finite subcover such that

$$X = \bigcup_{i=1}^n \bigcup_{j=1}^m B_{\alpha_i}^{\gamma_j} \subset \bigcup_{i=1}^n \mathcal{U}_{\alpha_i}$$

Thus,  $\{\mathcal{U}_{\alpha_i} \mid i = 1, 2, \dots, n\}$  is a finite subcover, we desired.

□

**Theorem 0.1.7.** Let  $X, Y$  are Topological Space. Then, TFAE:

- a)  $X \times Y$  is Compact.
- b)  $X$  and  $Y$  both are Compact.

*Proof.* a)  $\implies$  b) is clear since projection preserves Compactness.

b)  $\implies$  a) Let  $\mathcal{O} \stackrel{\text{def}}{=} \{U \times V \mid U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}$  be an Open cover of  $X \times Y$ .

Let  $x \in X$  fix. Then,  $\{x\} \times Y$  be a Compact, being  $\{x\} \times Y \cong Y$  by Homeomorphism given by Projection.

Then, there is a finite subcover of  $\mathcal{O}$  such that

$$\{x\} \times Y \subset \bigcup_{i=1}^{n_x} (U_i^x \times V_i^x)$$

Now, for each  $x \in X$ , define  $U^x \stackrel{\text{def}}{=} \bigcap_{i=1}^{n_x} U_i^x$ . Then,  $U^x$  is an open set containing  $x$ , and for any  $i = 1, 2, \dots, n_x$ ,  $U^x \subset U_i^x$ .

Since  $\{U^x \mid x \in X\}$  be an open cover of  $X$ , there is a finite subcover such that

$$X = \bigcup_{i=1}^m U^{x_i}$$

being  $X$  is Compact. Now,

$$X \times Y = \left( \bigcup_{i=1}^m U^{x_i} \right) \times Y = \bigcup_{i=1}^m (U^{x_i} \times Y) \subset \bigcup_{i=1}^m \bigcup_{j=1}^{n_{x_i}} (U_j^{x_i} \times V_j^{x_i})$$

Thus,  $\{U_j^{x_i} \times V_j^{x_i} \mid i = 1, 2, \dots, m, j = 1, 2, \dots, n_{x_i}\}$  be a finite subcover. □

#### Tube Lemma

Let  $X$  be a Topological Space, and  $Y$  is Compact Space.

Then, for product space  $X \times Y$ , and fixed  $x_0 \in X$ , following statement holds:

For any open  $N \subset X \times Y$  with  $\{x_0\} \times Y \subset N$ , there is an open  $W \in \mathcal{T}_X$  such that  $\{x_0\} \times Y \subset W \times Y \subset N$ .

*Proof.* Clearly,  $\{x_0\} \times Y$  compact, being  $\{x_0\} \times Y \simeq Y$ .

For any  $y \in Y$ ,  $(x_0, y) \in \{x_0\} \times Y \subset N$ , thus there exist opens  $U \in \mathcal{T}_X$  and  $V \in \mathcal{T}_Y$  such that  $(x_0, y) \in U \times V \subset N$ .

Now, Clearly  $\{U_y \times V_y \subset X \times Y \mid y \in Y\}$  be an open cover of  $\{x_0\} \times Y$ , thus there is a finite subcover such that

$$\{x_0\} \times Y \subset \bigcup_{i=1}^N (U_{y_i} \times V_{y_i}) \subset N$$

Set  $W = \bigcap_{i=1}^N U_{y_i}$ . Then, clearly  $x_0 \in W$ , and

Let  $(x, y) \in W \times Y$ . Then, since  $Y = \bigcup_{i=1}^n V_{y_i}$ , there is  $1 \leq k \leq n$  such that  $y \in V_{y_k}$ .

Thus,  $(x, y) \in U_{y_k} \times V_{y_k} \subset N$ , this implies  $W \times Y \subset N$ . □

**Theorem 0.1.8.** Let  $Y$  be a Compact Space. Then, the following statements are true, but their converses are false:

1. If  $X$  be a Lindelöf Space, then the product Topology  $X \times Y$  be a Lindelöf Space.
2. If  $X$  be a Countable Compact Space, then the product Topology  $X \times Y$  be a Countable Compact Space.

*Proof.* 1. Let  $\mathcal{O}$  be an open cover of  $X \times Y$ .

For any  $x \in X$ ,  $\{x\} \times Y$  is compact set, being  $\{x\} \times Y \simeq Y$ . Thus, there is a finite subcover of  $\mathcal{O}$  such that

$$\{x\} \times Y \subset \bigcup_{j=1}^{N_x} U_j^x \quad (U_j^x \in \mathcal{O})$$

Since Tube Lemma, there is an open  $W_x \in \mathcal{T}_X$  such that

$$\{x\} \times Y \subset W_x \times Y \subset \bigcup_{j=1}^{N_x} U_j^x$$

Meanwhile, since  $X$  is Lindelöf, therefore for an open cover  $\{W_x \mid x \in X\}$  there exists a Countable subcover such that

$$X \subset \bigcup_{i=1}^{\infty} W_{x_i}$$

Consequently,

$$X \times Y \subset \left( \bigcup_{i=1}^{\infty} W_{x_i} \right) \times Y \subset \bigcup_{i=1}^{\infty} (W_{x_i} \times Y) \subset \bigcup_{i=1}^{\infty} \bigcup_{j=1}^{N_{x_i}} U_j^{x_i}$$

Now,  $\{U_j^{x_i} \mid i \in \mathbb{N}, 1 \leq j \leq N_{x_i}\} \subset \mathcal{O}$  be a Countable Open Cover of  $X \times Y$ . □

*Proof.* 2. Let  $\{U_n \subset X \times Y \mid n \in \mathbb{N}\}$  be a Countable open cover of  $X \times Y$ . For each finite subset  $F \subset \mathbb{N}$ , define

$$V_F \stackrel{\text{def}}{=} \left\{ x \in X \mid \{x\} \times Y \subset \bigcup_{n \in F} U_n \right\}$$

Then  $V_F$  satisfies:

1)  $V_F$  is open: Let a finite subset  $F \subset \mathbb{N}$  fix. For each  $x \in V_F$ ,  $\{x\} \times Y \subset \bigcup_{n \in F} U_n$  by definition.

Then, there is an open  $W_x \in \mathcal{T}_X$  such that  $\{x\} \times Y \subset W_x \times Y \subset \bigcup_{n \in F} U_n$  by Tube Lemma.

Meanwhile,  $W_x \subset V_F$  because for all  $s \in W_x$ ,  $\{s\} \times Y \subset W_x \times Y \subset \bigcup_{n \in F} U_n$ , thus  $s \in V_F$ .

In summary, for any  $x \in V_F$ , there is an open  $W_x \in \mathcal{T}_X$  such that  $x \in W_x \subset V_F$ . Consequently,  $V_F$  is open of  $X$ .

2)  $\{V_F \mid F \subset \mathbb{N}, |F| < \infty\}$  is a Countable Open Cover of  $X$ :

Countability given by above set is collection of subsets of Countable set. Meanwhile,

For any  $x \in X$ , there is a finite subcover of  $\{U_n \mid n \in \mathbb{N}\}$  such that  $\{x\} \times Y \subset \bigcup_{n \in F} U_n$  where  $F$  finite.

That is,  $x \in V_F$ . Now, the open cover of  $X$ ,

$$\{V_{F_x} \mid x \in X\} \subset \{V_F \mid F \subset \mathbb{N}\}$$

at most Countable. Since  $X$  is Countably Compact Space, there is a finite subcover such that

$$X \subset \bigcup_{i=1}^N V_{F_i}$$

Consequently,

$$X \times Y \subset \left( \bigcup_{i=1}^N V_{F_i} \right) \times Y = \bigcup_{i=1}^N (V_{F_i} \times Y) \subset \bigcup_{i=1}^N \bigcup_{n \in F_i} U_n$$

That is,  $\{U_i \mid i = 1, 2, \dots, N, n \in F_i\}$  be a finite subcover. □